

|                                      | Column heading    | Parameter description                               | Unit               | Interval | Dataset name        | Type      | Comments                                                                                                                 | Data collection     |
|--------------------------------------|-------------------|-----------------------------------------------------|--------------------|----------|---------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|---------------------|
|                                      | Tout              | outside temperature                                 | °C                 | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Rhout             | outside Relative Humidity                           | %                  | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Iglob             | radiation                                           | W/m2               | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Windsp            | wind speed                                          | m/s                | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | RadSum            | radiation sum                                       | l/cm2              | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Winddir           | wind direction                                      | 0-360°             | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Rain              | rain (status 1=rain, 0=dry)                         | 1=rain, 0=dry      | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | PARout            | PAR weather measurement                             | µmol/m²/s          | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Pyrgeo            | heat emission: pyrgeometer                          | W/m2               | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | AbsHumOut         | absolute humidity content outside air               | g/m3               | 5 min    | meteo               | raw data  |                                                                                                                          | Weather station     |
|                                      | Time              | Timestamp 5 minute (Excel format)                   |                    |          |                     |           |                                                                                                                          | Weather station     |
| U6<br>U8<br>U12<br>U1<br>U5<br>Tp1pe | Tair              | Air temperature greenhouse                          | °C                 | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Measuring box       |
|                                      | RHum              | Relative humidity greenhouse                        | %                  | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Measuring box       |
|                                      | CO2air            | CO2 greenhouse                                      | ppm                | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Measuring box       |
|                                      | HumDef            | Humidity deficit                                    | g/m3               | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | VentLee           | Vent lee                                            | % (0-100)          | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | Ventwind          | Vent wind                                           | % (0-100)          | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | AssimLight        | Assimilation lighting                               | % (0-100)          | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | EnScr             | Energy curtain                                      | % (0-100)          | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | BlackScr          | Black-out curtain                                   | % (0-100)          | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | PipeLow           | Lower circuit Temperature                           | °C                 | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | PipeGrow          | Growth circuit Temperature                          | °C                 | 5 min    | Ghclimate           | raw data  |                                                                                                                          | Process computer    |
|                                      | GHTime            | Timestamp 5 minute (Excel format)                   |                    |          |                     |           |                                                                                                                          | Process computer    |
|                                      | Water             | Irrigation water supplied                           | L/m2               | Daily    | Irrigation          | Processed | Time between last and next irrigation turn                                                                               |                     |
|                                      | Drain             | Drainage water                                      | L/m2               | Daily    | Irrigation          | Processed |                                                                                                                          |                     |
|                                      | pH_Drain          | pH of drainage water                                | [-]                | Daily    | Irrigation          | Processed | Daily averages on 5 minute data                                                                                          | pH Probes           |
|                                      | EC_Drain          | EC of drainage water                                | dS/m               | Daily    | Irrigation          | Processed | Daily averages on 5 minute data                                                                                          | EC Probes           |
|                                      | Time              | Timestamp 1 day (Excel format)                      |                    |          |                     |           |                                                                                                                          |                     |
|                                      | CO2_VIP           | CO2 setpoint                                        | ppm                | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | HumDef_Vip        | humidity deficit setpoint                           | g/kg               | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | MinPipeLow_Vip    | net pipe minimum temperature setpoint               | °C                 | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | MinPipeGrow_Vip   | crop pipe minimum temperature setpoint              | °C                 | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | HeatTemp_VIP      | heating temperature setpoint                        | °C                 | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | VentLeew_VIP      | ventilation temperature lee side setpoint           | °C                 | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | VentLeewMin_VIP   | lee side window position minimum setpoint           | %                  | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | VentWind_VIP      | ventilation temperature wind side setpoint          | °C                 | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | WaterSupInt_VIP   | water supply interval time setpoint                 | minutes            | 5 min    | Vip                 | raw data  |                                                                                                                          | Process computer    |
|                                      | Time              | Timestamp 5 minute (Excel format)                   |                    |          |                     |           |                                                                                                                          |                     |
| U10                                  | Heating_Eenrgy    | Heating energy consumption (rail+ grow pipes)       | KWh/m2 day         | Daily    | ResourceConsumption | Processed | Energy sum from rail and growing pipe                                                                                    |                     |
|                                      | Electricity_Lamps | Electricity consumption (artificial light)          | KWh/m2 day         | Daily    | ResourceConsumption | Processed | From cumulative number of hours the lights were on, lamp efficiency (1.6µmol/J) and lamp intensity (187 µmol/m2 s)       |                     |
|                                      | CO2_dosage        | CO2 consumption                                     | Kg/m2 day          | Daily    | ResourceConsumption | Processed | From cumulative number of hours the valve was open and dosing capacity (0.0128 Kg/h m2)                                  |                     |
|                                      | net_water         | Water use (net of drainage)                         | L/m2 day           | Daily    | ResourceConsumption | Processed | Irrigation minus Drain                                                                                                   |                     |
|                                      | Labour            | Labour hours for crop management and harvest        | hours/day          | Daily    | ResourceConsumption | Processed | Number of hours per day per m2 corrected by a factor (0.42) to take into account the small-scale experimental greenhouse |                     |
|                                      | Time              | Timestamp 1 day (Excel format)                      |                    |          |                     |           |                                                                                                                          |                     |
|                                      | Prod_value_cum    | Production value (fruit number multiplied by price) | €                  | Daily    | Production          | Processed | Fruit number * cucumber price                                                                                            |                     |
|                                      | ProdA_cum         | Cumulative Production cucumber category A           | Kg/m2              | Daily    | Production          | Processed | Cumulative sum                                                                                                           | Manual registration |
|                                      | ProdB_cum         | Cumulative Production cucumber category B           | Kg/m2              | Daily    | Production          | Processed | Cumulative sum                                                                                                           | Manual registration |
|                                      | ProdA_num         | Cumulative number of cucumber category A            | number of fruits   | Daily    | Production          | Processed | Cumulative number sum                                                                                                    | Manual registration |
|                                      | ProdB_num         | Cumulative number of cucumber category B            | number of fruits   | Daily    | Production          | Processed | Cumulative number sum                                                                                                    | Manual registration |
|                                      | Total_Prod_cum    | Total cumulative production                         | Kg/m2              | Daily    | Production          | Processed | Sum class A and class B                                                                                                  | Manual registration |
|                                      | Time              | Timestamp 1 day (Excel format)                      |                    |          |                     |           |                                                                                                                          |                     |
|                                      | Fruit maintained  | % of fruit maintained on internode                  | %                  | week     | Crop Management     | raw       |                                                                                                                          |                     |
|                                      | Leaves maintained | % of leaves maintained on internode                 | %                  | week     | Crop Management     | raw       |                                                                                                                          |                     |
|                                      | Stem_Elong        | stem-elongation                                     | cm/week            | week     | Crop Management     | raw       |                                                                                                                          | Manual registration |
|                                      | FruitGrw          | fruit development time                              | days/fruit         | week     | Crop Management     | raw       |                                                                                                                          | Manual registration |
|                                      | LeafFormRate      | Leaf formation rate                                 | n_leaves/stem week | week     | Crop Management     | raw       |                                                                                                                          | Manual registration |
|                                      | N_leaves          | Cumulative number of leaves                         | n_leaves/stem      | week     | Crop Management     | raw       |                                                                                                                          | Manual registration |
|                                      | Weeks             | Timestamp week number (year 2018)                   |                    |          |                     |           |                                                                                                                          |                     |